

Phase 2: AI-Based Prediction and Optimization

Step 1: Generate Simulation Dataset

Run the Phase 1 model multiple times by varying the key parameters.

Input Parameters

Parameter	Sample Values
Particle size (nm)	20, 50, 100, 150, 200
Vessel pressure (mmHg)	15, 20, 25
Hydraulic conductivity	Low, Medium, High
Cellular uptake rate	0.02–0.10
Diffusion coefficient	Computed using Stokes–Einstein equation

Generate approximately **500–1000 simulations**.

For each simulation, store:

Particle Size Pressure Diffusion Uptake Penetration Depth Max Concentration

This becomes the AI dataset.

Step 2: Train Machine Learning Models

Split the dataset into:

- **80% Training**
- **20% Testing**

Train and compare:

- Random Forest Regressor
- XGBoost Regressor
- Decision Tree
- Neural Network (optional)

The model learns

Input Parameters → Penetration Depth

instead of solving the PDE every time.

Result 6

AI Prediction Accuracy

Plot

Actual Penetration Depth vs Predicted Penetration Depth

Evaluate using

- MAE
- RMSE
- R^2 Score

Expected

$R^2 > 0.95$

Step 3: Multi-Output Prediction

Instead of predicting only penetration depth,
predict simultaneously

- Penetration depth
 - Maximum concentration
 - Drug coverage
 - Drug delivery time
-

Result 7

Plot

Actual vs Predicted

Penetration Depth

Maximum Concentration

Drug Coverage

Compare prediction errors.

Step 4: Explainable AI

Use

SHAP (SHapley Additive exPlanations)

to identify which parameters have the greatest impact.

Result 8

Feature Importance

Example

Particle Size 

Pressure 

Diffusion 

Uptake 

Hydraulic Conductivity 

Students should explain

Particle size contributes most because it directly affects diffusivity.

Step 5: AI Optimization

Now ask

Which nanoparticle size gives maximum penetration?

Use the trained model to search for the optimal parameters.

Methods

- Bayesian Optimization
 - Genetic Algorithm
 - Grid Search
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Result 9

Optimal Parameters

Example

Parameter	Optimal Value
Particle Size	38 nm
Vessel Pressure	22 mmHg
Uptake Rate	0.04
Penetration Depth	7.4 mm

Step 6: Model Comparison

Compare the machine learning models.

Model	MAE	RMSE	R ²
Random Forest	0.15	0.20	0.95
XGBoost	0.08	0.12	0.99
Decision Tree	0.24	0.31	0.89

Choose the best-performing model.

Result 10

Performance Comparison

Plot

- Bar chart of MAE
- Bar chart of RMSE
- Bar chart of R^2

Students conclude

XGBoost achieved the highest prediction accuracy with the lowest prediction error.

Complete Phase 2 Workflow

Phase 1 Mathematical Model

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Generate Simulation Data

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Create AI Dataset

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Train ML Models

(Random Forest, XGBoost)

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Predict Drug Delivery

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Explain Predictions (SHAP)

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Optimize Nanoparticle Design



Compare AI Models

Final Outputs of Phase 2

Result Output

Result 6 AI model predicts penetration depth (Actual vs Predicted)

Result 7 Multi-output prediction (penetration, concentration, coverage)

Result 8 SHAP feature importance showing influential parameters

Result 9 Optimal nanoparticle size and transport parameters

Result 10 Comparison of AI models (MAE, RMSE, R^2)

Expected Deliverables

By the end of Phase 2, the students should have:

- A simulation dataset generated from the Phase 1 mathematical model.
- Trained AI models capable of predicting drug delivery outcomes.
- Explainable AI analysis identifying the most influential parameters.
- An optimized nanoparticle design based on AI predictions.
- A comparative study demonstrating the best-performing machine learning model.

This structure provides a clear transition from **physics-based simulation (Phase 1)** to **AI-assisted prediction and optimization (Phase 2)**, making the project suitable for publication in computational drug delivery and healthcare AI.